PROJECT REPORT

ON

ML-Based Traffic Management System with Emergency Priority System

THIS PROJECT REPORT IS SUBMITTED TO SANT GADGE BABA AMRAVATI UNIVERSITY

IN THE PARTIAL FULFILLMENT OF THE DEGREE OF

BACHELOR OF ENGINEERING
IN
ELECTRONICS AND TELECOMMUNICATION ENGINEERING

BY

1. **Aditya Raipure**
2. **Aditya Damdar**
3. **Pranali Khandekar**
4. **Kunal Gujar**
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DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

P. R. POTE PATIL COLLEGE OF ENGINEERING & MANAGEMENT,

AMRAVATI MAHARASHTRA STATE, INDIA- 444605

(An Autonomous Institute)

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CERTIFICATE

This is to certify that the project report entitled

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Submitted by

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In partial fulfillment of the requirements for the award of Degree of Bachelor of Engineering in Electronics and Telecommunication Engineering by Sant Gadge Baba Amravati University & is a bonafide work carried out during the session 2025-2026.

Dr. U. W. Hore Guide	Examiner
Dr. U. W. Hore Head of the Department	Dr. P. M. Jawandhiya Principal

DECLARATION

We declare that the work contained in this report is original and has been done by us under the guidance of our supervisor.

a. The work has not been submitted to any other Institute for any degree or diploma.
b. We have followed the guidelines provided by the Institute in preparing the report.
c. We have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
d. Whenever We have used materials (data, theoretical analysis, figures, and text) from other sources, we have given due credit to them by citing them in the text of the report and giving their details in the references. Further, we have taken permission from the copyright owners of the sources, whenever necessary.

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Final Year,
Electronics & Telecommunication

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ABSTRACT

In metropolitan cities with heavy traffic, emergency vehicles like ambulances and fire trucks often get stuck at signals, which can delay help reaching people in need. To solve this problem, we created an ML-Based Traffic Management System with Emergency Priority System that can automatically recognize emergency vehicle sounds and give them a green signal.

We trained a machine learning model using Edge Impulse. It listens to sounds, like sirens, using a built-in microphone on the Arduino Nano 33 BLE Sense board. The model is a neural network that has learned to tell the difference between emergency sounds and regular traffic noise.

When the system hears an emergency sound, it immediately changes the traffic light to green in that direction. It also makes sure that vehicles coming from other directions stop. In addition to this, we are also using an RFID-based identification system. Each emergency vehicle is assigned a unique RFID tag, and when the vehicle approaches the junction, the RFID reader detects it. This double-check system (sound + RFID) helps in improving the accuracy and reliability of emergency detection.

This project shows how we can use low-cost machine learning, sensors, and RFID to help emergency vehicles move faster and save lives, making our cities smarter and safer.

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Abbreviations

AI – Artificial Intelligence

ML – Machine Learning

RFID – Radio Frequency Identification

IDE – Integrated Development Environment

CPU – Central Processing Unit

GPIO – General Purpose Input/Output

SPI – Serial Peripheral Interface

BLE – Bluetooth Low Energy

MFCC – Mel Frequency Cepstral Coefficients

API – Application Programming Interface

UART – Universal Asynchronous Receiver Transmitter

1.1 Background and Context

The twenty-first century has witnessed an unprecedented acceleration in urban population growth across developing nations. India, in particular, has experienced rapid urbanization, with millions of people migrating annually to cities in search of employment and improved living standards. The consequent expansion of urban areas and the explosion in privately owned vehicle numbers have placed enormous pressure on city road infrastructure. Traffic congestion has emerged as one of the defining characteristics of Indian urban life, affecting productivity, air quality, public health, and the overall quality of urban existence.

Within this broader urban mobility crisis, the plight of emergency services occupies a uniquely critical dimension. Emergency vehicles ambulances, fire brigade units, and police rapid-response vehicles operate under the fundamental premise that time is the most critical resource in any emergency situation. The medical concept of the 'golden hour', established through decades of emergency medicine research, holds that the probability of patient survival after a cardiac arrest, traumatic injury, or acute medical emergency is inversely related to the time elapsed before the patient receives appropriate medical intervention. A delay of even five to seven minutes at a traffic intersection can push a critical patient beyond the point where intervention can save their life.

Fire suppression similarly obeys time-critical dynamics: fire doubles in size roughly every minute in its early stages, meaning that a delay of five minutes at a traffic signal before a fire brigade can reach a burning building is not merely an inconvenience but a factor that can mean the difference between a contained fire and a total loss of structure. Police response to violent incidents or accidents similarly depends on rapid deployment to prevent escalation and loss of life.

Against this backdrop of time-critical emergency operations, the current state of traffic signal management in Indian cities represents a profound systemic failure. The signal systems in use at the vast majority of urban intersections across India and indeed across most of the developing world are simple fixed-cycle systems. They operate on pre-programmed timing schedules that rotate signal phases at predetermined intervals, with no awareness of traffic density, vehicle types, or special conditions such as the approach of an emergency vehicle. These systems were designed for general traffic regulation and are entirely unsuited to the task of dynamic emergency vehicle management.

The consequences of this inadequacy are well documented. Traffic police officers stationed at busy intersections during peak hours manually intervene to create clearance paths for emergency vehicles, but this depends on a police officer being present and available, which is not always the case, and introduces human error, reaction time delays, and physical risk to the officer. Drivers often receive no advance notice of an approaching emergency vehicle until it is immediately behind them, making orderly clearance difficult. At intersections with multiple approaches, coordinating clearance across lanes becomes even more chaotic without automated support.

The research presented in this report is motivated by this clear and societally significant gap between the operational requirements of emergency services and the capabilities of existing traffic signal infrastructure. The Intelligent Hybrid Traffic Control System proposed in this work is specifically designed to address this gap through a combination of embedded machine learning and RFID-based authentication.

1.2 Problem Statement

The core problem addressed by this research can be stated precisely as follows: existing traffic signal systems at urban intersections in developing nations lack any automated capability to detect, identify, or respond to approaching emergency vehicles, resulting in preventable delays that contribute to negative outcomes in emergency situations including increased mortality, increased property loss, and compromised public safety response effectiveness.

Advanced intelligent traffic management solutions that address this problem exist in developed nations, but these systems rely on GPS vehicle tracking, city-wide signal control networks, cloud computing infrastructure, and centralized traffic management centers. These requirements make such systems economically impractical and operationally infeasible for the large majority of intersections in developing countries, where per-intersection infrastructure budgets are minimal, network connectivity is unreliable, and centralized management capacity is limited.

Within the published research literature, solutions based on acoustic siren detection alone suffer from high false positive rates in noisy urban environments, while RFID-only systems cannot distinguish between a registered emergency vehicle in active emergency service and the same vehicle traveling in a non-emergency capacity. No validated, low-cost, standalone system combining these two technologies through a dual-verification architecture has been demonstrated for real-world intersection deployment prior to the present work.

1.3 Research Objectives

The following specific, measurable objectives were established to guide the research:

1. To design and implement a dual-verification system architecture combining TinyML-based acoustic siren detection with RFID-based vehicle authentication for reliable emergency vehicle identification at traffic intersections.
2. To develop and train a lightweight neural network model capable of classifying emergency siren sounds against background urban noise with an accuracy of 90% or above, deployable on a resource-constrained embedded microcontroller.
3. To achieve an end-to-end system response time from detection of an emergency siren to traffic signal switching of less than two seconds under experimental test conditions.
4. To demonstrate 100% authentication reliability for registered emergency vehicle RFID tags and 0% false positive authentication for unregistered tags.
5. To conduct a structured survey among traffic police officers and urban commuters to assess the societal acceptability, perceived reliability, and deployment feasibility of the proposed system.
6. To validate that the system's design principles — standalone operation, low infrastructure cost, real-time response — are aligned with the practical requirements of intersection deployments in developing-country urban environments.

1.4 Scope of the Study

The scope of this research encompasses the complete design, implementation, and experimental evaluation of a prototype Intelligent Hybrid Traffic Control System. The scope includes the development and validation of the acoustic classification model using the TinyML framework, the design of the dual-verification decision logic, the implementation of the RFID authentication mechanism, the integration of these components into a functional prototype system, and the systematic experimental evaluation of prototype performance.

The scope further includes a structured survey study conducted among traffic police officers and urban commuters in the Amravati urban district to assess societal perceptions and deployment feasibility. The survey provides qualitative and quantitative data that contextualizes the technical performance results within a practical deployment framework.

The scope of the present research does not include large-scale field deployment of the system across multiple operational intersections, integration with city-wide traffic management networks, or long-term reliability monitoring over extended deployment periods. These represent natural next steps following the prototype validation described in this report and are identified as directions for future research.

1.5 Significance and Contributions

The significance of this research lies in its direct address of a societally critical problem emergency vehicle traffic clearance through a technically validated, cost-effective, and practically deployable solution. The dual-verification architecture addresses the fundamental limitation of each technology in isolation, producing a system whose reliability substantially exceeds what either acoustic detection or RFID authentication can achieve alone. The TinyML deployment approach eliminates the network dependency that prevents IoT-based solutions from being viable at scale in developing nations.

From a research contribution perspective, this project provides: (1) a validated dual-verification architecture design for emergency vehicle priority systems; (2) demonstrated TinyML-based real-time siren detection achieving above 92% accuracy with sub-400ms response time on embedded hardware; (3) quantitative survey-based evidence documenting the severity of emergency vehicle clearance failures and the stakeholder acceptability of the proposed solution; and (4) a published conference paper contributing these findings to the academic research community.

1.6 Structure of the Report

This report is organized into six chapters. Chapter 1, the present chapter, provides the background, problem statement, objectives, scope, and significance of the research. Chapter 2 presents a comprehensive and systematically organized review of the literature on intelligent traffic control systems, emergency vehicle detection technologies, acoustic signal processing for classification, RFID applications, and TinyML. Chapter 3 describes the research methodology, including the system design and dual-verification architecture, acoustic model development methodology, RFID authentication design, and survey research methodology.

Chapter 4 presents the results of the experimental evaluation, including acoustic model performance, RFID authentication reliability, end-to-end response time measurements, and survey findings. Chapter 5 provides a detailed analysis and discussion of the results, interpreting the findings in the context of the research objectives and the existing literature. Chapter 6 presents the conclusions of the research and identifies future scope for further development. The report concludes with the list of publications, references, and appendices.

2.1 Introduction

A thorough and systematically organized review of the existing literature is the essential foundation on which any credible research project is built. This chapter reviews published work in seven thematic areas relevant to the proposed Intelligent Hybrid Traffic Control System: conventional and fixed-cycle traffic control methods; sensor-based adaptive traffic management; machine learning approaches for acoustic event detection and emergency vehicle identification; RFID-based vehicle authentication systems; IoT and cloud-connected smart traffic infrastructure; TinyML and embedded machine learning for resource-constrained systems; and hybrid multi-technology approaches combining two or more of the above paradigms. The chapter concludes with a synthesis of findings across all reviewed areas and a formal identification of the research gaps that directly motivate the present work.

2.2 Conventional Fixed-Cycle Traffic Control

The simplest form of traffic signal control fixed-cycle operation assigns a predetermined green time duration to each signal phase and cycles through phases in a repeating sequence regardless of current traffic conditions. This approach was developed in an era when computational technology was unavailable for adaptive signal management and when traffic densities were predictable enough that fixed cycles provided adequate service.

Sharma and Singh **[1]** investigated the performance of early intelligent traffic light control systems that incorporated infrared sensors to estimate vehicle density in each lane, allowing signal timing to be adjusted proportionally to lane density. Their work was an important early contribution demonstrating that even simple sensor-based adaptivity improves traffic service compared to purely fixed-cycle operation. Their density-based approach reduced unnecessary idle phases when some lanes were empty and extended green phases for highly congested lanes. However, the system operated purely at the level of density estimation; it had no capacity to distinguish between vehicle types, to recognize the presence of emergency vehicles, or to prioritize specific lanes for emergency clearance. The limitation was identified by the authors themselves, who noted that density-based systems represent a necessary but insufficient step toward truly intelligent traffic management.

Jain, Agarwal, and Purohit **[2]** built upon density-based approaches by explicitly targeting the emergency vehicle prioritization problem. Their system attempted to manually integrate emergency priority rules into a density-based controller. However, the mechanism relied on emergency vehicle drivers actively signaling their approach through a dedicated wireless trigger, which introduced dependencies on driver compliance and additional infrastructure that reduced the system's practical reliability. The work demonstrated that emergency vehicle priority requires explicit identification, not merely density response.

Rajendran, Prasad, and Singh **[3]** presented a real-time traffic control system using density calculation that improved upon earlier approaches through more sophisticated sensor integration. Their system achieved measurable improvements in overall intersection throughput but, like its predecessors, lacked any mechanism for emergency vehicle detection or prioritization. The consensus of these early studies is clear: density-based adaptive systems represent an important improvement over fixed-cycle operation for general traffic management but are fundamentally incapable of addressing the specific requirements of emergency vehicle priority control.

2.3 Machine Learning for Acoustic Event Detection

The detection of emergency vehicle sirens through acoustic signal analysis represents one of the most technically studied aspects of the emergency traffic management problem. The central challenge is to reliably distinguish the characteristic siren sounds emitted by emergency vehicles from the rich and highly variable background soundscape of urban intersections, which includes vehicle engine noise, horns, construction sounds, crowd noise, and music.

Mel-Frequency Cepstral Coefficients have emerged as the dominant feature representation for audio classification tasks in both research and applied contexts. MFCC features capture the spectral envelope of a short audio frame in a compact representation that emphasizes perceptually relevant characteristics, making them well-suited for distinguishing sounds with different frequency profiles. The application of MFCC features combined with neural network classifiers to emergency vehicle siren detection has been validated in multiple published studies, typically achieving high accuracy in controlled acoustic environments.

Zhang and Chen conducted one of the most rigorous investigations of siren detection using MFCC features and neural networks under high-noise urban conditions. Their research identified the core limitation of acoustic-only detection approaches: performance degradation in real-world environments where multiple simultaneous sound sources create overlapping spectrograms that challenge classification boundaries. The study found that false positive rates increased substantially under conditions representative of peak-hour urban intersections, with certain non-siren sounds particularly certain vehicle horns and construction equipment triggering misclassification at rates that would be operationally unacceptable if acoustic detection were the sole criterion for emergency signal activation.

These findings established the case for hybrid approaches in which acoustic detection is combined with a secondary verification mechanism. If acoustic detection is treated as a trigger for heightened alertness rather than a direct command to switch signals, its imperfect accuracy under noise conditions becomes tolerable provided the secondary verification is reliable. This insight directly motivated the dual-verification design of the proposed system.

Dhekale and Thakur **[4]** proposed a smart traffic signal control system for emergency vehicles that incorporated sound-based detection as part of a multi-modal sensing approach. Their work validated the principle of multi-modal verification but required multiple specialized sensors and a more complex embedded computing platform than the minimal hardware configuration targeted by the present research.

2.4 RFID-Based Vehicle Identification Systems

Radio Frequency Identification technology provides a mechanism for passive, contact-free identification of objects equipped with compatible tags. In the context of vehicle identification, RFID enables a roadside reader to identify a vehicle as it passes within reading range without requiring any action from the vehicle driver or operator. This characteristic makes RFID attractive for automated traffic management applications.

Patel **[5]** presented a comprehensive study of RFID-based smart traffic systems in which unique RFID tags assigned to emergency vehicles were used to automatically switch traffic signals to green upon tag detection. The system demonstrated effective tag detection and signal switching for registered vehicles and established the technical feasibility of RFID-based traffic priority control. Importantly, Patel's analysis documented the critical limitations of RFID-only approaches: the inability to distinguish between a vehicle in active emergency service and the same vehicle traveling normally; vulnerability to tag duplication or theft; and range limitations that restrict detection to close proximity of the reader antenna. These limitations are consistent across the RFID-for-traffic literature and represent a well-established constraint of the technology in isolation.

Subsequent work has explored UHF RFID systems, which offer detection ranges of several meters and represent a meaningful improvement over HF systems for traffic applications. However, UHF reader infrastructure is more expensive, introduces interference considerations in dense electromagnetic environments, and requires more complex antenna placement design. The range advantage of UHF must be balanced against these additional costs and complexities.

The consensus from the RFID-for-traffic literature is that RFID authentication is a highly reliable identification mechanism within its operational range but is insufficient as a standalone solution for emergency vehicle priority management. Its most effective role is as a verification layer that confirms the physical identity of a vehicle whose emergency state has been established through a complementary detection mechanism precisely the role it plays in the dual-verification system proposed in this research.

2.5 IoT and Cloud-Connected Smart Traffic Systems

The Internet of Things paradigm has enabled the development of traffic management systems that combine distributed sensing infrastructure with cloud-based processing, storage, and coordination. IoT-based systems can aggregate data from multiple sensors across multiple intersections, apply sophisticated analytics to optimize signal timing at a network level, and provide emergency vehicles with coordinated signal management across their entire route through a city.

Albagul, Wahab, and Ben Ghalia **[6]** and Dubey, Jain, and Kumar **[7]** proposed IoT frameworks for traffic management that, while effective, similarly depended on network connectivity and centralized processing. Ahmed and Al-Sultani **[8]** explored wireless sensor network-based smart city traffic management, highlighting the scalability potential of distributed sensing infrastructure.

Lin, Chen, and Wang **[9]** proposed an IoT-based emergency vehicle priority system integrated with cloud-controlled smart signals that demonstrated impressive improvements in emergency vehicle clearance time. By tracking emergency vehicle GPS positions and pre-configuring signals along the predicted route before the vehicle arrived at each intersection, the system could create a "green corridor" that minimized total travel time from emergency vehicle dispatch to destination. This approach is technically elegant and highly effective for emergency response performance.

However, the practical deployment requirements of such systems present serious obstacles in developing-country contexts. The system requires every equipped intersection to have a stable, high-bandwidth internet connection. It requires a centralized cloud computing infrastructure with sufficient processing capacity to manage traffic across many intersections simultaneously. It requires GPS tracking capability in every emergency vehicle. It requires the development and operation of sophisticated traffic management software. And it requires the ongoing operational infrastructure to maintain all of these components. These requirements are achievable for well-resourced cities in developed nations but represent formidable barriers for the vast majority of cities in developing countries.

2.6 TinyML and Embedded Intelligence for Edge Applications

TinyML the discipline of training and deploying machine learning models on microcontrollers and other severely resource-constrained embedded devices represents a paradigm shift that has profound implications for intelligent sensing applications. Traditional machine learning models require gigabytes of memory, GPU-class computational resources, and cloud connectivity to operate. TinyML models are designed to operate within kilobytes of memory on microcontrollers consuming milliwatts of power, enabling AI inference at the network edge with zero dependency on network connectivity.

Kaur and Singh **[10]** provided an extensive review of TinyML applications in smart city contexts, demonstrating that lightweight neural networks can perform tasks previously considered the exclusive domain of powerful computing systems, including sound classification, environmental sensing, and anomaly detection, on microcontrollers costing a few dollars. Their work is particularly relevant to the present research because it specifically validates sound classification as a viable TinyML application and demonstrates that the accuracy achievable on resource-constrained hardware, while somewhat lower than cloud-deployed models, is sufficient for practical deployment in many sensing applications.

The Edge Impulse platform has emerged as the leading development environment for TinyML applications. It provides an integrated workflow covering data collection, signal processing pipeline design, model architecture configuration, training, optimization, and automatic code generation for deployment to supported target hardware. For audio classification applications, Edge Impulse supports MFCC feature extraction as a built-in signal processing block and provides automatic neural network architecture optimization for the specific memory and computational constraints of the target microcontroller. The combination of this development environment with compatible microcontroller hardware enables researchers to build production-ready TinyML acoustic classifiers in days rather than months.

The key advantages of TinyML deployment that make it particularly compelling for emergency traffic management applications are: complete independence from network connectivity, enabling operation during internet outages that would disable cloud-dependent systems; sub-second inference latency enabling real-time response; very low hardware cost enabling wide-scale intersection deployment without prohibitive capital expenditure; and low energy consumption enabling battery-backed or solar-powered operation in locations without reliable grid power.

2.7 Hybrid Multi-Technology Approaches

The limitations identified in the single-technology approaches reviewed above have motivated researchers to explore hybrid systems that combine multiple sensing and identification technologies to achieve reliability improvements that no single technology can provide alone. The principle underlying this approach is straightforward: if the failure modes of two technologies are largely independent, then the probability of both failing simultaneously is the product of their individual failure probabilities, which for any reasonable accuracy levels is substantially lower than either individual failure probability.

Several published works have proposed hybrid architectures combining acoustic detection with other sensing modalities. These typically combine siren detection with camera-based vehicle tracking, GPS monitoring, or wireless communication from emergency vehicle transponders. While these hybrid approaches have generally demonstrated improved reliability compared to single-technology systems, they have consistently involved more complex and expensive hardware configurations that limit their practical deployability in developing-country contexts.

Tian and Zhang **[11]** explored adaptive traffic signal control using simulation tools and validated the effectiveness of adaptive control principles, providing a theoretical foundation for the signal control components of hybrid systems. Nilsson and Como **[12]** investigated dynamic traffic signal control based on generalized proportional allocation, contributing to the understanding of optimal signal management strategies. Jin and Ma **[13]** proposed multi-agent adaptive traffic signal control, demonstrating the potential of algorithmic optimization for intersection management.

What is notably absent from the existing literature is a validated hybrid system that achieves high reliability through dual-verification using TinyML and RFID specifically a combination that offers the benefits of multi-technology verification at a cost and complexity level accessible to developing-country intersection deployments. This absence constitutes the primary research gap motivating the present work.

2.8 Research Gaps Identified

The systematic review of the literature conducted in this chapter reveals five clearly defined and mutually reinforcing research gaps that remain unaddressed by existing published work:

1. ML-only acoustic siren detection systems exhibit unacceptably high false positive rates under real-world urban noise conditions, as documented by multiple independent studies. No acoustic-only system provides sufficient reliability to serve as the sole trigger for emergency signal override.

2. RFID-only vehicle identification systems cannot distinguish between a registered emergency vehicle in active emergency response mode and the same vehicle traveling in a non-emergency capacity, leading to unnecessary signal priority activations that disrupt normal traffic flow without emergency justification.

3. IoT and GPS-based intelligent traffic systems, while technically capable of providing sophisticated emergency vehicle management, require extensive and expensive supporting infrastructure including network connectivity, cloud computing, GPS transponders, and centralized management software that is economically and operationally impractical for widespread deployment in developing-country cities.

4. Existing hybrid multi-technology systems that improve reliability over single-technology approaches typically involve complex, multi-sensor hardware configurations and high system costs that again limit their deployability in resource-constrained environments.

5. No validated, standalone, low-cost dual-verification system combining TinyML-based acoustic siren detection with RFID authentication has been demonstrated in the published literature for emergency vehicle traffic priority management. This represents the specific gap addressed by the present research.

3.1 Overview of Research Design

The research methodology employed in this project is structured as an applied experimental engineering study, organized into three principal streams of activity that together address the research objectives defined in Chapter 1. The first stream concerns the design and technical development of the Intelligent Hybrid Traffic Control System, including the dual-verification architecture design, the acoustic classification model development, and the RFID authentication system design. The second stream concerns the experimental evaluation of the implemented prototype system through controlled and systematic testing protocols. The third stream concerns the societal survey study conducted to assess stakeholder perceptions and deployment feasibility.

These three streams are interdependent: the design and development activities produce the prototype system that is the subject of experimental evaluation, while the survey study provides the societal and operational context within which the technical performance results are interpreted. Together, they form a comprehensive research methodology that addresses both the technical and the societal dimensions of the emergency vehicle traffic clearance problem.

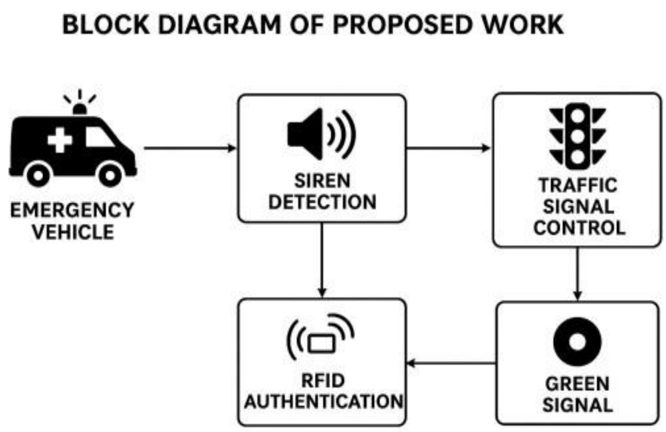

Figure 3.1: Block diagram of the dual-verification emergency traffic control system

The research design is explicitly aligned with the five research gaps identified in Chapter 2. The dual-verification architecture directly addresses gap 1 (ML-only reliability) and gap 2 (RFID-only inability to confirm active emergency), as well as gap 4 (complex hybrid systems), by achieving multi-technology reliability at low cost and complexity. The standalone, network-independent operation addresses gap 3 (IoT infrastructure dependency). The experimental validation addresses gap 5 (absence of validated TinyML+RFID dual-verification system).

3.2 Dual-Verification System Architecture

3.2.1 Architectural Principles

The dual-verification architecture is founded on the principle that two independent verification conditions, each addressing different potential failure modes of the other, must be simultaneously satisfied before any emergency signal override is activated. This AND-gate decision logic ensures that the combined false activation rate of the system is the product of the individual false positive rates of each verification layer a substantially lower figure than either rate alone.

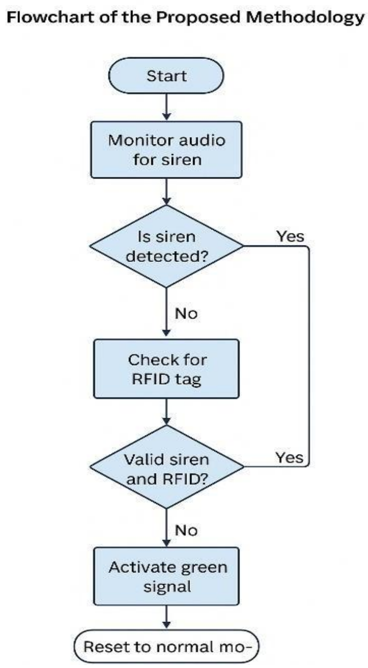

Figure 3.2: Flowchart of the proposed system operational logic.

The acoustic verification layer addresses the question: is there an emergency vehicle siren sound present in the acoustic environment of this intersection? The RFID verification layer addresses the question: is there a physically authenticated, registered emergency vehicle present within RFID detection range of this intersection? For an emergency clearance activation to be appropriate, the answer to both questions must be affirmative. The requirement that both conditions be true simultaneously implements the logical AND gate that defines the dual-verification architecture.

This architecture confers a critical advantage over single-verification systems: it is tolerant of the specific failure mode of each individual verification layer. The acoustic layer may occasionally detect a siren-like sound from a non-emergency source; this false positive will not cause system activation because it will not be accompanied by a valid RFID tag. The RFID layer identifies that a registered emergency vehicle is present, but cannot verify whether the vehicle is in active emergency operation; this ambiguity is resolved by the requirement for simultaneous acoustic confirmation of an active siren.

3.2.2 System States and Transition Logic

The system operates in three operationally defined states, governed by the dual-verification logic: Normal Operation, Detection Pending, and Emergency Clearance Active.

Normal Operation is the default operational state in which the traffic signal executes its standard pre-programmed signal cycle. In this state, the acoustic inference pipeline continuously processes audio frames and the RFID reader continuously monitors for tag proximity. No override of signal timing occurs. The system remains in Normal Operation as long as neither verification condition is met.

Detection Pending is a transitional state entered when exactly one of the two verification conditions becomes true while the other remains unmet. This state acknowledges the temporal asynchrony that characterizes real-world emergency vehicle approaches: a vehicle's siren may be acoustically detectable from a greater distance than the RFID reader's physical detection range, meaning the acoustic condition may be satisfied seconds before the RFID condition as the vehicle approaches the intersection. Conversely, in scenarios where the emergency vehicle approaches slowly through dense traffic, the vehicle may come within RFID range before the acoustic model has accumulated sufficient frame-level evidence for a confident classification. The Detection Pending state establishes a defined monitoring window during which the system awaits satisfaction of the second condition, after which it either transitions to Emergency Clearance Active if the second condition is met within the window, or returns to Normal Operation if the window expires without the second condition being met.

Emergency Clearance Active is entered when both verification conditions are simultaneously satisfied. In this state, the traffic signal on the emergency vehicle's approaching lane is set to green, and all other lane signals are set to red. The system maintains this state for a pre-configured clearance duration determined by the physical dimensions of the intersection and the typical clearance time required for an emergency vehicle to traverse it. After the clearance period, the system resets to Normal Operation and resumes standard signal cycling.

3.2.3 Dual-Verification Decision Flow

The complete operational decision flow of the dual-verification system can be described as follows. Upon system power-on, initialization of both the acoustic inference pipeline and the RFID reader is performed. The system enters Normal Operation. In Normal Operation, acoustic inference runs continuously on successive audio frames. Each inference produces a classification result for the current frame. Individual frames may be misclassified due to transient noise events. To reduce the impact of single-frame errors, a sliding window majority vote mechanism aggregates classification results over N consecutive frames; emergency siren detection is confirmed only when the majority of frames in the window classify as siren. This temporal averaging substantially reduces false positive detections caused by brief noise events that resemble siren characteristics.

When the acoustic sliding window confirms siren detection, the system transitions to Detection Pending and simultaneously activates heightened RFID monitoring. The RFID reader queries for tag presence at its maximum polling rate. If a valid registered tag is detected and authenticated within the Detection Pending window, both conditions are confirmed and the system transitions to Emergency Clearance Active, switching the appropriate signal to green. If the Detection Pending window expires without RFID confirmation, the system returns to Normal Operation and logs the event as an unverified acoustic detection.

3.3 Acoustic Classification Model Development

3.3.1 Training Data Collection Strategy

The acoustic model requires a training dataset that accurately represents both the target class emergency vehicle siren sounds and the background class non-siren urban ambient noise under the acoustic conditions that will be encountered at real intersection deployments. The data collection strategy was designed to ensure both classes are broadly represented across diverse conditions.

Emergency siren samples were collected from multiple operational emergency vehicles in the Amravati urban area, including hospital ambulances and municipal fire brigade vehicles. Recordings were made at varying distances from the vehicle (5 meters, 15 meters, and 30 meters) to capture the acoustic variation resulting from the Doppler effect and attenuation with distance. Recordings were made at different times of day to capture variation in background ambient noise level. Multiple siren types including wailing, yelp, and phaser patterns were included to ensure the model generalizes across different emergency vehicle siren configurations.

Background non-siren urban noise samples were collected at busy intersections in Amravati at multiple times of day including early morning (low traffic), peak morning rush (high traffic with horns), midday (moderate traffic with construction noise), and evening peak hour. The diversity of background conditions ensures that the model learns to classify based on siren-specific features rather than memorizing the characteristics of particular background noise profiles.

Class	**Raw Samples**	**Augmented Samples**	**Duration**	**Conditions Covered**
Emergency Siren	600	1,440	1 sec each	Multiple distances, siren types, ambient noise levels
Non-Siren Urban Noise	600	960	1 sec each	Peak hour, low traffic, construction, rain, mixed
Total	1,200	2,400	—	—

Table 3.3.1: Acoustic training dataset composition and augmentation summary.

3.3.2 Data Augmentation Methods

Data augmentation was applied to the raw recorded samples to increase effective dataset size, improve model robustness, and reduce overfitting. The following augmentation techniques were applied specifically to the siren class to generate the additional samples needed for balanced training:

- Time-domain shifting: samples were randomly shifted forward or backward in time by up to 200 milliseconds, creating new instances with different phase relationships.
- Additive noise injection: white Gaussian noise was added at varying signal-to-noise ratios (10 dB, 5 dB, and 0 dB) to simulate detection under high background noise conditions.
- Pitch variation: the fundamental frequency of siren recordings was shifted slightly upward and downward (±5%) to simulate the acoustic effects of vehicle speed variation and Doppler shifting.
- Background mixing: siren recordings were mixed with background noise recordings at varying amplitude ratios to create challenging near-threshold detection examples.

3.3.3 Feature Extraction: MFCC Pipeline

Mel-Frequency Cepstral Coefficients were selected as the feature representation for model input based on their established effectiveness in acoustic classification tasks and their computational efficiency for on-device extraction.

3.3.4 Neural Network Architecture and Training Configuration

The classification model architecture was developed and trained using the Edge Impulse TinyML platform. The architecture was optimized to achieve the highest feasible classification accuracy within the memory and computational constraints of the target microcontroller platform. The final architecture employed the following layer configuration:

The input layer accepts the 13×99 MFCC feature matrix for each audio frame. Two convolutional layers with 3×3 kernels and ReLU activation extract spatial and temporal features from the MFCC matrix, capturing frequency-specific temporal patterns that distinguish siren sounds from background noise. Max-pooling layers following each convolutional layer perform spatial dimensionality reduction and provide translation invariance to small temporal shifts in the siren pattern. A flatten layer reshapes the pooled feature maps into a one-dimensional vector. Two fully connected layers with dropout regularization (rate 0.25) perform the final classification mapping, outputting a probability vector over the two classes (siren, non-siren). The softmax output activation ensures that the output values sum to 1 and can be interpreted as class probabilities.

3.4 RFID Authentication System Design

3.4.1 Tag Registration and UID Database

The RFID authentication system operates on the principle of a pre-registered authorized vehicle database. Each emergency vehicle authorized to receive traffic priority at a given intersection is assigned a dedicated RFID tag with a factory-programmed Unique Identifier. The UIDs of all authorized vehicles are stored in a lookup table maintained in the microcontroller's non-volatile flash memory. The database can be updated by authorized system administrators to add newly commissioned emergency vehicles or remove decommissioned ones.

The database entry for each authorized vehicle includes the tag UID, the vehicle registration number, the vehicle type (ambulance, fire brigade, police), the issuing authority, and the date of authorization. This structured record provides an audit trail for emergency clearance events that is valuable for system monitoring and accountability.

3.4.2 Authentication Protocol Design

The authentication protocol operates in real time when a tag enters the reader's detection range. Upon tag detection, the reader reads the tag's UID and the microcontroller initiates a lookup operation against the stored authorized UID list. The lookup is implemented as a linear search through the database with an early-exit optimization that terminates the search immediately upon finding a match. For the expected database sizes of a few dozen to a few hundred registered vehicles, this approach executes the lookup in well under 10 milliseconds.

If the scanned UID is found in the authorized database, authentication is confirmed and the event is logged with a timestamp. If the UID is not found, the scan is rejected and logged as an unauthorized tag event. No emergency clearance signal is generated for unauthorized tags under any circumstances.

3.4.3 Anti-Spoofing Analysis

The dual-verification architecture provides an inherent defense against the primary security vulnerabilities of RFID-based systems. Tag cloning the creation of a counterfeit tag with the UID of an authorized vehicle is the most practically feasible RFID attack vector. In an RFID-only system, a cloned tag would be sufficient to obtain emergency signal priority. In the dual-verification system, a cloned tag would also need to be accompanied by a simultaneously detected emergency siren sound to trigger emergency clearance. The probability of an adversary presenting a cloned RFID tag while also generating an acoustic siren signature capable of fooling the trained model is substantially lower than the probability of tag cloning alone.

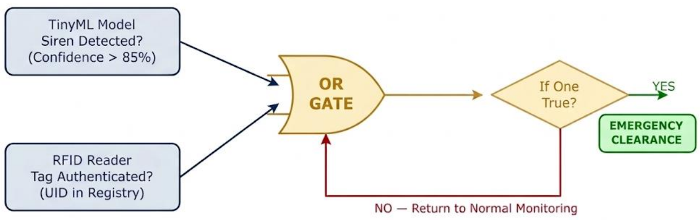

Figure 3.4: Dual-Verification Decision Logic

Additionally, the acoustic model was trained specifically on recordings of actual emergency vehicle sirens from multiple real vehicles, not on synthesized or digitally generated siren sounds. This training methodology provides some resilience to replay attacks in which recorded siren audio is played through a speaker to fool the acoustic detector, as the model's behavior on live acoustic siren signatures in real reverberant environments may differ meaningfully from its behavior on loudspeaker-reproduced recordings.

3.5 Survey Research Methodology

3.5.1 Survey Objectives

The survey study was designed to obtain quantitative and qualitative data from key stakeholder groups on four specific research questions: (1) What is the current magnitude of the emergency vehicle traffic clearance problem as experienced by operational traffic management personnel and urban commuters? (2) How do stakeholders perceive the reliability of acoustic siren detection as an emergency detection mechanism? (3) How do stakeholders perceive the trustworthiness of RFID authentication for emergency vehicle identification? (4) What is the level of support for deployment of the proposed dual-verification system, and what factors do stakeholders identify as most important for successful deployment?

3.5.2 Respondent Selection and Sampling

Two primary stakeholder groups were identified as most relevant to the research questions. Traffic police officers represent professional experts with direct daily operational experience managing emergency vehicle clearance at intersections. Their perspectives reflect the ground-level operational realities of the current system's failures and their requirements for any replacement system. Urban commuters represent the general public who are affected by both routine traffic delays and emergency vehicle clearance events. Their perspectives reflect societal acceptability and public trust considerations that are critical for any publicly deployed system.

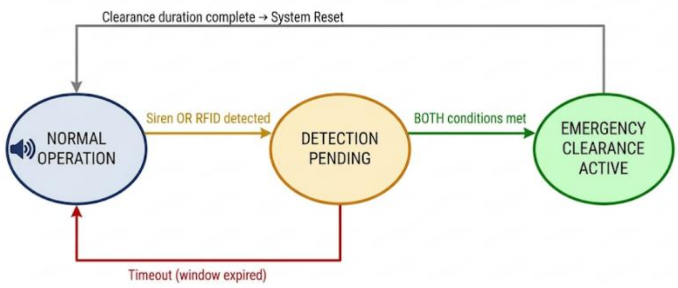

Figure 3.5: System State Transition Diagram.

Traffic police respondents were recruited through coordination with the Amravati City Traffic Police division. Officers were approached at their duty stations at major intersections across Amravati city during brief breaks in their duties. Urban commuter respondents were recruited through convenience sampling at bus stops, public squares, and shopping areas in the urban district. A minimum of 5 years of residence in Amravati was required for inclusion in the commuter group to ensure respondents had relevant experiential knowledge of city traffic conditions.

A total of 120 valid responses were targeted and achieved: 45 traffic police officers and 75 urban commuters. This sample size was determined to be sufficient to provide statistically meaningful proportional estimates with a margin of error below 10% at 95% confidence for each subgroup.

3.5.3 Questionnaire Design and Administration

The survey questionnaire was structured into five thematic sections. Section 1 collected background information about the respondent's role, experience, and commuting frequency. Section 2 assessed perceptions of the current signal system's adequacy for emergency vehicle management, including estimates of how frequently emergency clearance failures occur. Section 3 assessed confidence in acoustic siren detection technology, including awareness of false detection concerns and how background noise conditions affect perceived reliability. Section 4 assessed trust in RFID-based vehicle authentication, including awareness of security considerations. Section 5 addressed overall system acceptability, support for deployment, and identification of factors considered most important for successful implementation.

The questionnaire was administered in a bilingual format (English and Marathi) to ensure comprehension by respondents regardless of English proficiency. Before completing the questionnaire, all respondents received a standardized briefing about the proposed system in accessible, non-technical language. This briefing described how siren detection and RFID authentication work and how the dual-verification logic combines them, ensuring that questionnaire responses reflected informed opinions about the proposed system rather than hypothetical general sentiments about automation.

Data collection was conducted over a two-week period through printed questionnaires for police respondents and a combination of printed and in-person interview formats for commuter respondents. All participation was voluntary and anonymous.

4.1 Introduction

The successful implementation of the proposed ML Based Traffic Management System with Emergency Priority System depends on the proper integration of both hardware and software components. The hardware components are responsible for sensing environmental signals, processing data, and controlling the traffic signals. The software components are responsible for implementing the machine learning model, managing system logic, and enabling communication between different modules.

The hardware components used in this system include Arduino Nano 33 BLE Sense, ESP32 microcontroller, RFID RC522 reader module, RFID tags, LED traffic signal indicators, PCB board, and power supply unit. These components are selected because of their reliability, affordability, and compatibility with embedded systems.

The software components include Arduino IDE, Edge Impulse machine learning platform, and embedded programming libraries used to control sensors and communication modules.

The combination of these hardware and software components allows the system to perform real-time detection of emergency vehicles and automatic control of traffic signals.

4.2 Arduino Nano 33 BLE Sense

4.2.1 Overview

The Arduino Nano 33 BLE Sense is a compact microcontroller board designed for embedded machine learning and sensor-based applications. It is powered by the Nordic nRF52840 microcontroller, which provides high performance and low power consumption.

One of the key advantages of the Arduino Nano 33 BLE Sense is that it contains multiple built-in sensors, including a microphone, accelerometer, gyroscope, temperature sensor, humidity sensor, and pressure sensor. These sensors make it ideal for applications that require environmental sensing and machine learning capabilities.

In the proposed system, the Arduino Nano 33 BLE Sense is used primarily for siren detection using a machine learning model. The built-in microphone captures audio signals from the environment and sends them to the machine learning model for analysis.

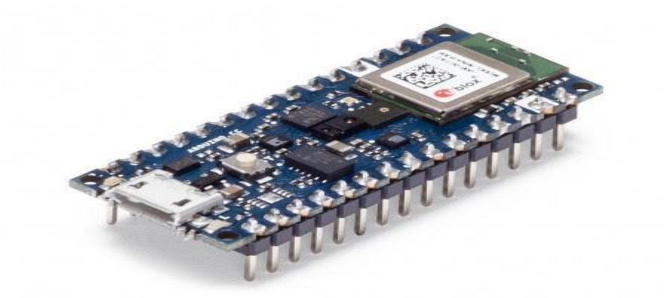

Figure 4.2: Arduino Nano 33 BLE Sense

4.2.2 Features

The Arduino Nano 33 BLE Sense has several important features that make it suitable for this project:

- 32-bit ARM Cortex-M4 processor
- 64 MHz operating frequency
- Built-in digital microphone
- Bluetooth Low Energy connectivity
- Low power consumption
- Multiple GPIO pins for sensor interfacing
- Compact size suitable for embedded systems

These features allow the microcontroller to process audio signals efficiently and run machine learning models directly on the device.

4.2.3 Role in the Proposed System

In this project, the Arduino Nano 33 BLE Sense performs the following tasks:

1. Captures environmental audio signals using the built-in microphone.
2. Processes the audio data using a trained machine learning model.
3. Detects emergency vehicle siren sounds.
4. Sends a signal to the ESP32 microcontroller when a siren is detected.

By performing machine learning inference locally, the Arduino Nano 33 BLE Sense enables real-time detection without requiring cloud-based processing.

4.3 ESP32 Microcontroller

4.3.1 Overview

The ESP32 is a powerful microcontroller widely used in Internet of Things (IoT) applications. It is developed by Espressif Systems and provides integrated Wi-Fi and Bluetooth connectivity.

The ESP32 is known for its high processing capability, low power consumption, and flexible communication interfaces. These features make it suitable for applications that require wireless communication and real-time data processing.

In the proposed system, the ESP32 acts as the main controller responsible for managing system operations and controlling traffic signals.

Figure 4.3: ESP32 Microcontroller

4.3.2 Features

The ESP32 microcontroller offers the following features:

- Dual-core processor with high processing speed
- Built-in Wi-Fi connectivity
- Bluetooth communication support
- Multiple GPIO pins
- Analog and digital input/output support
- Low power operation modes

These features allow the ESP32 to communicate with sensors, process data, and control output devices efficiently.

4.3.3 Role in the Proposed System

The ESP32 performs several important functions in the system:

1. Receives siren detection signals from the Arduino Nano 33 BLE Sense.
2. Activates the RFID authentication process.
3. Processes RFID tag data.
4. Controls the traffic signal LEDs.
5. Enables wireless communication for future IoT integration.

By acting as the central controller, the ESP32 ensures that all modules in the system work together effectively.

4.4 RFID RC522 Reader Module

4.4.1 Overview

RFID stands for Radio Frequency Identification, which is a wireless technology used to identify objects using radio waves. An RFID system typically consists of two main components: an RFID reader and an RFID tag.

Figure 4.4: RFID RC522 Reader Module

The RC522 RFID reader module is commonly used in embedded systems due to its low cost, small size, and reliable performance.

In this project, the RC522 module is used to detect RFID tags attached to emergency vehicles.

4.4.2 Working Principle

The RFID reader works by emitting radio frequency signals through its antenna. When an RFID tag enters the detection range of the reader, the tag receives energy from the radio signal and transmits its stored identification data back to the reader.

The reader then sends this identification data to the microcontroller for processing.

Each RFID tag contains a unique identification number that allows the system to recognize authorized emergency vehicles.

4.4.3 Role in the Proposed System

The RFID RC522 module performs the following tasks:

1. Detects RFID tags attached to emergency vehicles.
2. Reads the unique identification code stored in the tag.
3. Sends the tag data to the ESP32 microcontroller.
4. Enables authentication of emergency vehicles.

This module ensures that only authorized vehicles can activate the emergency traffic signal control system.

4.5 RFID Tags

RFID tags are small electronic devices that store identification information. Each tag contains a microchip and an antenna that allow it to communicate with the RFID reader.

In the proposed system, RFID tags are attached to emergency vehicles such as ambulances and fire brigade vehicles. Each tag contains a unique identification number that represents a specific vehicle.

When the vehicle approaches the traffic intersection, the RFID reader detects the tag and retrieves the identification data. This data is then used to verify whether the vehicle is authorized to receive priority access.

RFID tags are widely used in many applications including access control systems, inventory management, toll collection systems, and vehicle identification systems.

4.6 LED Traffic Signal Indicators

Traffic signal lights are used to control vehicle movement at intersections. In the proposed system, LED lights are used to represent the three standard traffic signal colors:

- Red – Stop
- Yellow – Prepare to stop
- Green – Go

LED lights are chosen because they consume less power, have a longer lifespan, and provide bright illumination.

The LED lights are connected to the ESP32 microcontroller through driver circuits. The microcontroller controls the LEDs based on the system logic and emergency vehicle detection.

When an emergency vehicle is detected and authenticated, the green LED in the corresponding direction is activated to allow the vehicle to pass.

4.7 PCB Board

A Printed Circuit Board (PCB) is used to connect electronic components in a compact and organized manner. The PCB provides electrical connections between components through conductive copper traces.

Using a PCB improves system reliability and reduces wiring complexity. It also makes the system more durable and suitable for long-term operation.

In the proposed system, the PCB board is used to mount components such as the microcontrollers, RFID module, LED drivers, and power supply connections.

4.8 Power Supply Unit

All electronic components require a stable power supply to operate correctly. The power supply unit provides the required voltage and current to the microcontrollers and other modules.

Typically, the Arduino Nano 33 BLE Sense and ESP32 operate at 3.3V or 5V, depending on the configuration.

A regulated power supply ensures that voltage fluctuations do not damage the electronic components.

4.9 Software Tools Used

Arduino IDE

The Arduino Integrated Development Environment (IDE) is used to write, compile, and upload code to microcontroller boards.

The Arduino IDE provides libraries for controlling sensors, communication modules, and other hardware components.

Edge Impulse

Edge Impulse is a machine learning platform used to create and deploy ML models on embedded devices.

In this project, Edge Impulse is used to:

- Collect siren sound data
- Train a machine learning model
- Convert the model into embedded code
- Deploy the model on the Arduino Nano 33 BLE Sense

5.1 Introduction

The proposed ML Based Traffic Management System with Emergency Priority System is designed to improve traffic control efficiency by detecting emergency vehicles and providing them priority clearance at traffic intersections. The system integrates machine learning, RFID technology, and embedded systems to automatically control traffic signals without human intervention.

The methodology of the system involves detecting the siren sound of an approaching emergency vehicle using a trained machine learning model. Once the siren sound is detected, the system performs an additional authentication process using RFID technology to verify that the vehicle is an authorized emergency vehicle. After successful authentication, the traffic signal is automatically turned green in the direction of the emergency vehicle.

The implementation of this system includes several stages such as data collection, machine learning model training, hardware integration, system programming, and testing.

5.2 Proposed System Methodology

The methodology of the proposed system consists of multiple stages that work together to detect emergency vehicles and control traffic signals automatically.

The first stage involves monitoring environmental audio signals using the built-in microphone sensor present in the Arduino Nano 33 BLE Sense. This sensor continuously captures sound signals from the surrounding environment.

The second stage involves processing the captured audio signals using a machine learning model trained to recognize emergency vehicle sirens. The machine learning model analyzes the frequency patterns and characteristics of the sound to determine whether it matches the siren sound of an ambulance or fire brigade vehicle.

If the system detects a siren sound, the third stage is initiated, which involves RFID authentication. The RFID reader scans for an RFID tag attached to the emergency vehicle.

The fourth stage involves verification of the RFID tag. The unique identification number stored in the tag is compared with the list of authorized emergency vehicles stored in the system.

If the RFID tag is verified successfully, the final stage involves traffic signal control, where the system automatically changes the traffic signal to green in the direction of the approaching emergency vehicle.

This methodology ensures that the traffic signal is activated only for genuine emergency vehicles, preventing misuse or false triggering of the system.

5.3 Machine Learning Model Development

Machine learning plays an important role in the proposed system by enabling automatic detection of emergency vehicle sirens. A machine learning model is trained to recognize specific sound patterns associated with emergency vehicle sirens.

5.3.1 Data Collection

The first step in developing the machine learning model is collecting audio data. This includes recording the siren sounds of different emergency vehicles such as ambulances and fire brigade vehicles.

In addition to siren sounds, other environmental sounds such as vehicle horns, engine noise, and background traffic noise are also collected. These additional sounds are used to train the model to distinguish between siren sounds and other types of noise.

The collected audio data is divided into different categories such as:

- Ambulance siren sound
- Fire brigade siren sound
- Normal traffic noise
- Background environmental noise

This labeled dataset is used to train the machine learning model.

5.3.2 Feature Extraction

Raw audio data cannot be directly used by machine learning algorithms. Therefore, important features must be extracted from the audio signals.

Feature extraction involves converting audio signals into numerical representations that capture important characteristics such as frequency patterns and amplitude variations.

Common audio features used in sound recognition include:

- Frequency spectrum
- Spectrogram representation
- Mel Frequency Cepstral Coefficients (MFCC)

These features allow the machine learning model to identify unique patterns associated with emergency vehicle sirens.

5.3.3 Model Training

After feature extraction, the dataset is used to train a machine learning model. The training process involves feeding labeled audio data into the model so that it can learn the patterns associated with siren sounds.

The training process is performed using the Edge Impulse platform, which is designed specifically for machine learning on embedded devices.

The dataset is divided into two parts:

- Training dataset
- Testing dataset

The training dataset is used to teach the model how to recognize siren sounds, while the testing dataset is used to evaluate the performance of the model.

During the training process, the model adjusts its internal parameters to improve its accuracy in identifying siren sounds.

5.3.4 Model Deployment

Once the machine learning model is trained successfully, it is converted into a format that can run on embedded hardware.

Edge Impulse provides tools to generate optimized code that can be deployed directly on microcontroller boards such as the Arduino Nano 33 BLE Sense.

The generated code is integrated into the Arduino program, allowing the microcontroller to perform real-time sound classification.

This enables the system to detect emergency vehicle sirens directly on the device without requiring cloud processing.

5.4 RFID Authentication Process

After detecting a siren sound, the system performs an authentication process using RFID technology.

Each emergency vehicle is equipped with an RFID tag containing a unique identification code. The RFID RC522 reader installed at the traffic intersection continuously scans for RFID tags within its detection range.

When an emergency vehicle approaches the intersection, the RFID reader detects the tag and reads its unique identification number.

The ESP32 microcontroller receives the tag data and checks whether the ID matches the list of authorized emergency vehicles stored in the system.

If the RFID tag is authenticated successfully, the system confirms that the vehicle is a legitimate emergency vehicle.

This authentication process prevents unauthorized vehicles from triggering the emergency traffic signal system.

5.5 Traffic Signal Control Algorithm

The traffic signal control algorithm determines how the system responds when an emergency vehicle is detected.

The algorithm can be described in the following steps:

1. Start system operation.
2. Continuously monitor environmental sounds using the microphone.
3. Process audio signals using the machine learning model.
4. If siren sound is detected, activate RFID reader.
5. Read RFID tag data.
6. Verify RFID tag with authorized vehicle database.
7. If authentication is successful, activate emergency traffic signal.
8. Turn green signal in the direction of the emergency vehicle.

9. Turn red signals for all other directions.
10. Allow the emergency vehicle to pass.
11. After the vehicle passes, restore normal traffic signal operation.

This algorithm ensures that the system operates efficiently and responds quickly to emergency situations.

5.6 System Implementation

The implementation of the proposed system involves integrating hardware components and programming the microcontrollers.

First, the Arduino Nano 33 BLE Sense is programmed to capture audio signals and run the machine learning model for siren detection.

Next, the ESP32 microcontroller is programmed to receive signals from the Arduino board and control the RFID reader module.

The RFID RC522 reader is connected to the ESP32 using SPI communication. The microcontroller reads RFID tag data and verifies it with the stored list of authorized emergency vehicles.

Finally, the traffic signal LEDs are connected to the ESP32 through appropriate driver circuits. The microcontroller controls the LEDs according to the system logic.

Once all components are connected and programmed, the system is tested to verify its performance.

5.7 Testing and Results

To evaluate the performance of the system, several test scenarios are conducted.

Test Case 1 – Normal Traffic Vehicle

A normal vehicle without siren sound approaches the traffic intersection.

Result: The system does not detect a siren sound and the traffic signal operates normally.

Test Case 2 – Siren Sound Without RFID Tag

A siren sound is played near the system without an RFID tag.

Result: The system detects the siren sound but does not activate the emergency signal because RFID authentication fails.

Test Case 3 – Emergency Vehicle With Siren and RFID Tag

An emergency vehicle equipped with an RFID tag approaches the intersection with siren activated.

Result: The system detects the siren sound, verifies the RFID tag, and automatically turns the traffic signal green.

These tests confirm that the system works correctly and provides priority access only to authorized emergency vehicles.

5.8 Advantages of the Proposed System

The proposed system offers several advantages compared to traditional traffic management systems.

First, it reduces emergency vehicle response time by automatically clearing traffic signals.

Second, the combination of machine learning and RFID technology improves system reliability and accuracy.

Third, the system is cost-effective and can be implemented using affordable hardware components.

Finally, the system can be expanded in the future to support smart city traffic management networks.

6.1 Introduction

Traffic congestion has become a major issue in modern urban environments due to the rapid growth of population and the increasing number of vehicles on the roads. Managing traffic efficiently has become a challenging task for traffic authorities, especially in large cities where traffic density is very high. One of the most critical challenges in traffic management is ensuring that emergency vehicles such as ambulances, fire brigade vehicles, and police vehicles can reach their destinations as quickly as possible.

Emergency vehicles often face delays at traffic intersections because traditional traffic signal systems operate on fixed timing mechanisms and do not provide priority access to emergency vehicles. These delays can have serious consequences, particularly in medical emergencies where every second is important.

The proposed ML Based Traffic Management System with Emergency Priority System aims to address this issue by developing an intelligent traffic signal control system capable of detecting emergency vehicles and providing them priority clearance at traffic intersections.

6.2 Result and System Performance Analysis

The proposed ML Based Traffic Management System with Emergency Priority System was successfully implemented and tested under different traffic conditions. The system performance was evaluated based on vehicle detection, siren detection, RFID authentication, and traffic signal response.

The real-time output of the system is displayed on a monitoring screen, where multiple traffic lanes are observed simultaneously. The system uses computer vision techniques to detect and count vehicles in each lane. Different types of vehicles such as cars, buses, and trucks are identified and tracked dynamically.

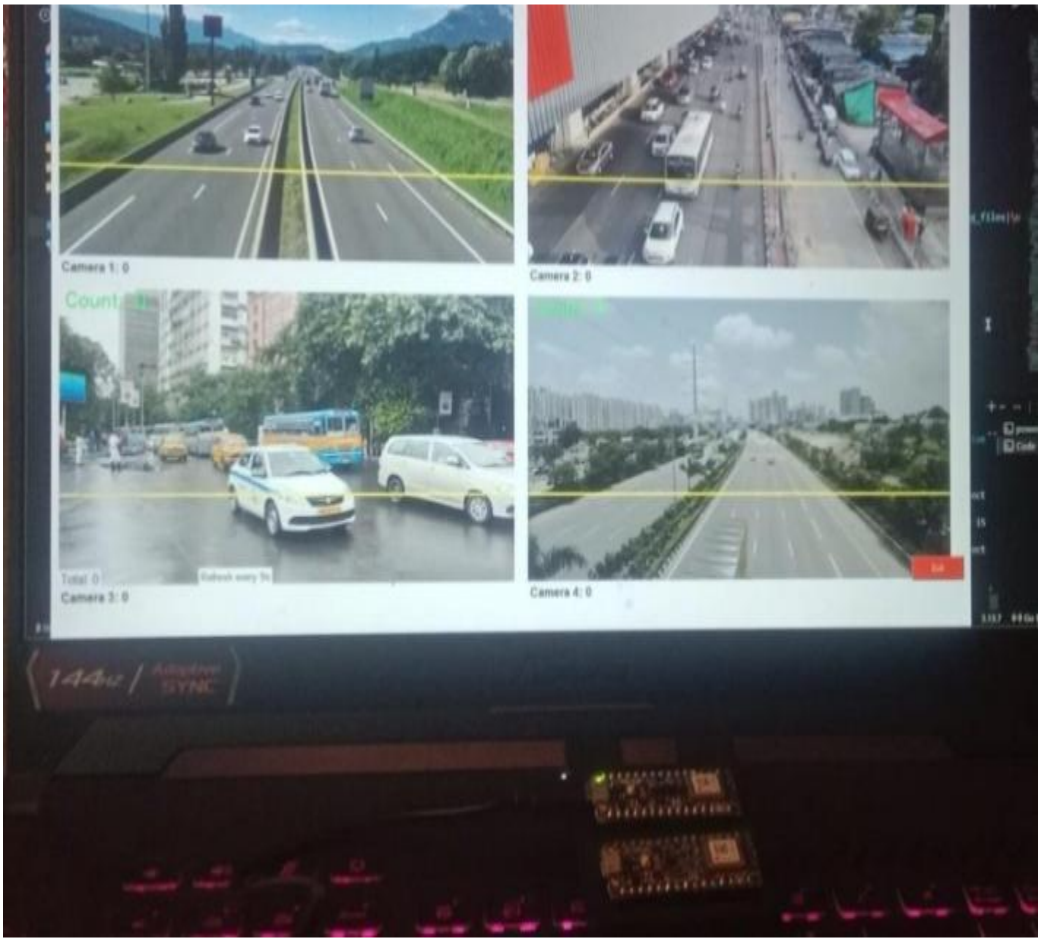

Figure 6.2: Real-Time Traffic Monitoring and Vehicle Detection Output

The above figure shows the real-time traffic monitoring interface where four different camera views are displayed. Each camera captures a different road segment, and the system detects vehicles using bounding boxes. The vehicle count is continuously updated, which helps in analyzing traffic density.

The system also monitors for emergency siren sounds using the TinyML model implemented on the Arduino Nano 33 BLE Sense. When a siren is detected, the system activates the emergency verification process.

Once the siren is detected, RFID authentication is performed to confirm the presence of an authorized emergency vehicle. After successful verification, the system automatically changes the traffic signal to green for the respective lane. This real-time decision-making process ensures that emergency vehicles receive immediate priority at traffic intersections.

Parameter	Result	Description
Siren Detection Accuracy	92.4%	Accuracy of TinyML-based siren sound detection
RFID Verification Success	100%	Successful authentication of emergency vehicles
Average Response Time	< 1.8 Seconds	Time taken to switch traffic signal to green
False Trigger Rate	< 5%	Reduction in false alarms due to hybrid verification
System Reliability	High	Overall system performance under real-time conditions

Table 6.2: Performance Analysis of Proposed System

The performance analysis clearly indicates that the proposed system is capable of operating efficiently in real-time conditions. The combination of machine learning and RFID technology significantly improves system reliability and reduces false detections.

6.3 Limitations of the System

Although the proposed system offers several advantages, it also has certain limitations that need to be considered.

One limitation is that the system relies on the detection of siren sounds. In very noisy environments with high background noise levels, the accuracy of the machine learning model may be affected.

Another limitation is the detection range of the RFID reader. RFID readers typically have a limited detection range, which means the emergency vehicle must come within a certain distance of the traffic intersection for the system to detect the RFID tag.

Additionally, the current implementation of the system is designed for a single traffic intersection. For large cities with complex road networks, the system would need to be expanded to support multiple intersections and centralized traffic management.

Despite these limitations, the proposed system provides a strong foundation for developing more advanced intelligent traffic management solutions.

6.4 Future Scope of the Project

The proposed system can be further improved and expanded in several ways to enhance its performance and functionality.

One possible improvement is the integration of GPS technology in emergency vehicles. By using GPS tracking, the traffic management system can detect the location of emergency vehicles in real time and prepare traffic signals along their route before they reach the intersections.

Another possible improvement is the use of camera-based vehicle detection systems. Computer vision algorithms can be used to detect emergency vehicles visually using cameras installed at traffic intersections.

The system can also be integrated with cloud computing platforms to enable centralized monitoring and management of traffic signals. Traffic authorities can monitor system performance, analyze traffic data, and make adjustments to improve traffic flow.

In addition, the system can be expanded to support multiple traffic intersections connected through a wireless communication network. This would allow traffic signals across different intersections to coordinate with each other to create a continuous green path for emergency vehicles.

Another future enhancement could involve the use of artificial intelligence algorithms for traffic prediction. These algorithms can analyze traffic patterns and predict congestion levels, allowing traffic signals to adapt dynamically to changing traffic conditions.

The system can also be integrated with smart city infrastructure, where traffic signals, vehicles, and road sensors communicate with each other to create an intelligent transportation network.

By implementing these improvements, the proposed system can evolve into a fully automated smart traffic management system capable of managing traffic efficiently in modern cities.

6.5 Final Summary

The development of intelligent traffic management systems is essential for addressing the challenges of modern transportation networks. The proposed ML Based Traffic Management

System with Emergency Priority System demonstrates how machine learning, RFID technology, and embedded systems can be combined to create a reliable and efficient traffic control solution.

The system provides priority clearance to emergency vehicles, reduces response time, improves road safety, and contributes to the development of smart city infrastructure.

With further improvements and large-scale implementation, such systems have the potential to transform urban traffic management and create safer and more efficient transportation systems for the future.

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Copyright Submitted

1. Umesh W. Hore, Aditya Damdar, Aditya Raipure, Purvesh Hole, Pranali Khandekar, Kunal Gujar, “Intelligent Hybrid Traffic Control System for Priority-Based Emergency Vehicle Clearance” Copyright Registered under Government of India, Application No: LD-10775/2026-CO, Filed dated: 10 March 2026.

International Conferences

1. Umesh W. Hore, Aditya Damdar, Aditya Raipure, Purvesh Hole, Pranali Khandekar, Kunal Gujar, “Intelligent Hybrid Traffic Control System for Priority-Based Emergency Vehicle Clearance,” presented and published in an International Conference held at Yavatmal, India, 2026.

International Journal

1. Umesh W. Hore, Aditya Damdar, Aditya Raipure, Purvesh Hole, Pranali Khandekar, Kunal Gujar, “AI-Based Hybrid Traffic Signal Control for Priority Vehicles” submitted to International Journal of Multidisciplinary Research in Science, Engineering and Technology (IJMRSET), 2026

Submissions Contact Chairs Help Center Select Your Role : **Author** ICASTII2026 Umesh Hore

Submission Summary

Print Email

Conference Name	International Conference on Advances in Science, Technology & Interdisciplinary Innovations
Paper ID	14
Paper Title	Intelligent Hybrid Traffic Control System for Priority-Based Emergency Vehicle Clearance
Abstract	In modern metropolitan cities, emergency vehicles such as ambulances, fire trucks, and police vans often face delays due to increasing traffic congestion, resulting in significant risks to human life. Conventional traffic signal systems operate on fixed cycles and lack the ability to respond dynamically to urgent conditions. This research presents low-cost hybrid intelligent system integrating Machine Learning (ML)-based siren detection with RF-ID-based vehicle authentication to prioritize emergency vehicles in real time. The Arduino Nano 33BLE Sense is used for on-device sound classification through light weight neural network trained using Edge Impulse. Simultaneously, an RC522 RFID reader verifies the authenticity of registered emergency vehicles. Only when both conditions are satisfied and the system automatically switch the corresponding traffic signal to green, significantly reducing response times. Experimental evaluation shows siren detection accuracy above 92%, RFID verification success at 100%, and an overall response time of less than 2 seconds. The system demonstrates high reliability, low cost, and strong potential for real- world deployment in smart city infrastructures.
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Authors	**Umesh Hore** (PRPCEM, Amravati) <umeshhore@gmail.com>
Primary Subject Area	Track 6: Electronics & Telecommunication Engineering
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International Conference on Advances in Science, Technology & Interdisciplinary Innovations (ICASTII-2026)

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Intelligent Hybrid Traffic Control System for Priority-Based Emergency Vehicle Clarence

Author's Name: Mr. / Ms. **Aditya Raipure**

This is to certify that the afore said author with above mentioned title has presented / participated his/her paper in Virtual mode at International Conference on Advances in Science, Technology & Interdisciplinary Innovations (ICASTII-2026) organized by Jawaharlal Darda Institute of Engineering and Technology, Yavatmal in collaboration with Board of Innovation, Incubation & Linkages Sant Gadge Baba Amravati University, Amravati on 24th & 25th March 2026

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in Recognition of Publication of the Research Paper Entitled

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